Cover Letter

A Self-Contained Real-Time Detection Pipeline and an Operating-Point Replay Protocol Applied to Turkish Broadband Stations

H. Oğuz Bolat,Sibel Çimen,Bülent Bolat,Tülay Yıldırım

Dear Editors-in-Chief,

Please find enclosed the manuscript “A Self-Contained Real-Time Detection Pipeline and an Operating-Point Replay Protocol Applied to Turkish Broadband Stations”, which we submit for exclusive consideration as a research article in Computers & Geosciences. We confirm that the submission follows the journal’s requirements and includes every item of its submission checklist.

The manuscript contributes in both dimensions that the journal asks for. Computationally, it presents ayzek, a self-contained C++23 pipeline for the detection stage of earthquake early warning. Its hand-written inference reproduces the PyTorch networks to within floating-point rounding, its replays of archived data are deterministic, and the same build runs on a Raspberry Pi 4. It also presents a replay protocol, implemented as a reproducible workflow, that measures detectors at their operating point with chance correction, pre-registered held-out sequences, paired tests and catalog-quiet data. Geoscientifically, it measures how a learned detector, two classical triggers and two published models behave on continuous records of Turkish broadband stations. At comparable unmatched alarms the learned detector detected 36.4 of 66 held-out events after chance correction against 21.7 for STA/LTA, and on a second held-out sequence the difference was not significant. Causal PhaseNet matched the learned detector, and on 102.8 h of catalog-quiet data the quietest STA/LTA settings raised 5 to 29 times as many alarms as the quietest learned ones. On a 728-day record, two detectors with nearly equal ROC-AUC differed by a factor of 1.6 in recall at one false alarm per day.

The source code, model weights, public test datasets and reproduction scripts are in a public repository under the GNU General Public License, with the details listed in the section “Code availability”. The AFAD waveforms cannot be redistributed, and public KOERI datasets reproduce the principal results.

The use of generative AI tools during this work is declared in the manuscript, and the authors reviewed and edited all content and take full responsibility for it. The manuscript has not been published and is not under consideration elsewhere, all authors approve its submission, and they declare no competing interests.

Sincerely,

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Highlights

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- Self-contained C++ detection pipeline reproduces PyTorch networks to rounding
- Replay protocol scores detectors at their operating point with chance correction
- Learned detectors find more small events than STA/LTA at equal unmatched alarms
- Quietest STA/LTA settings raised 5–29 times the alarms of learned ones on quiet data
- Detectors with near-equal ROC-AUC differed 1.6-fold in recall at one alarm a day

A Self-Contained Real-Time Detection Pipeline and an Operating-Point Replay Protocol Applied to Turkish Broadband Stations

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ABSTRACT

Earthquake early warning (EEW) requires detectors that run continuously on every station, where false alarms accumulate over hours without seismicity, yet learned detectors are mostly evaluated on curated windows. We present ayzek, a self-contained C++23 detection pipeline whose networks reproduce their PyTorch counterparts to within floating-point rounding and whose replays of archived data are deterministic, and a replay protocol that measures detectors at their operating point, with chance correction, pre-registered held-out sequences, paired tests and catalog-quiet data. Applied to Turkish broadband stations, the protocol compares a convolutional–recurrent detector with a recursive short-term average/long-term average (STA/LTA) trigger, the FilterPicker multi-band picker and the published EQTransformer and PhaseNet models, all run through the same pipeline. At comparable unmatched alarms the learned detector detected 36.4 of 66 events after correction for chance matches against 21.7 for STA/LTA on the first held-out sequence (difference 14.7, 95% interval 5.9–23.7), and 35.1 of 56 against 29.1 on the second, where the difference is not significant. Causal PhaseNet matched the learned detector on both, which indicates that the gain is a property of learned detection rather than of this detector. On 102.8 h without cataloged seismicity, at equal detections on the tuning data, the quietest STA/LTA settings raised 5 to 29 times as many alarms as the quietest learned ones, and FilterPicker matched the learned detector on events only at 55 times its alarm rate. On a 728-day record at one station, two detectors with nearly equal ROC-AUC differed by a factor of 1.6 in recall at one false alarm per day. The pipeline runs unchanged on a Raspberry Pi 4, and its code, weights, public test datasets and reproduction scripts are openly available.

CRediT authorship contribution statement

H. Oğuz Bolat: Conceptualization, Methodology, Software, Investigation, Formal analysis, Data curation, Visualization, Writing – original draft. **Sibel Çimen:** Writing – review and editing. **Bülent Bolat:** Writing – review and editing, Supervision. **Tülay Yıldırım:** Writing – review and editing, Supervision.

1. Introduction

Earthquake early warning (EEW) systems run a detector continuously on every station, and the P travel time to the stations needed to declare an event bounds the warning they can give (Allen et al., 2009). Over hours without seismicity, the detector's false alarms accumulate. Learned detectors, however, are mostly evaluated on curated windows with balanced classes, where a metric such as the area under the receiver operating characteristic curve (ROC-AUC) says little about the false-alarm rate at the operating point.

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This study contributes in both computational and geoscientific terms. Its computational contribution is twofold. The first part is ayzek, a self-contained C++23 pipeline from miniSEED decoding to the network alarm, with hand-written inference and no machine-learning runtime. Its networks are verified against their PyTorch counterparts, its replays of archived data are deterministic, and the same build runs on a Raspberry Pi 4. The second part is a replay protocol, implemented as a reproducible workflow, that measures detection at the operating point. Trigger settings are chosen on tuning data by rules stated in advance and then tested on held-out sequences, one of them chosen and committed before its data were extracted. Chance matches with the catalog are corrected by shifting the catalog in time, only the stations that raised an alarm are matched, detectors are compared with paired tests, and false alarms are counted on data without cataloged seismicity.

The geoscientific contribution is a measurement of how learned and classical detectors behave on continuous seismic records from Turkish broadband stations. The pipeline's convolutional–recurrent detector is compared with two classical detectors, a recursive STA/LTA trigger and FilterPicker (Lomax et al., 2012), and with two published learned models, EQTransformer (Mousavi et al., 2020) and PhaseNet (Zhu and Beroza, 2019), to test whether any gain over the classical trigger is specific to our detector. The comparison quantifies the false-alarm rates of these detectors on catalog-quiet data and shows on a 728-day record that ROC-AUC can rank detectors differently from their recall at operational alarm budgets. The study concerns the detection stage, which is the first stage of an alert. Magnitude estimation is outside its scope, and an alarm here is a network detection rather than a warning that an operational system could issue. All results come from replays of archived data from one network.

2. Background

Operational EEW systems are evaluated for latency as well as accuracy. The testing platform of ShakeAlert (Kohler et al., 2018) replays archived waveforms through the complete system (Cochran et al., 2018), which is the practice that the protocol of this study follows. Its detection stages are tuned against false as well as missed events, as for ElarmS-3 (Chung et al., 2019), and the alerts of the public system are reported in operation (Kohler et al., 2020). In the Marmara region, the Istanbul Earthquake Rapid Response and Early Warning System raises an alarm when filtered peak acceleration or cumulative absolute velocity exceeds a threshold at three or more strong-motion stations within 10 s (Alcik et al., 2009).

Classical detection relies on characteristic functions of the waveform, such as the ratio of short- to long-term averages (Allen, 1978), whose sensitivity to its parameters is well documented (Withers et al., 1998). Multi-band pickers designed for early warning, such as FilterPicker (Lomax et al., 2012), remain in operational use because they are inexpensive and interpretable and time an onset to the sample. Their weakness is specificity, because an amplitude criterion that is sensitive enough for small earthquakes is just as sensitive to impulsive cultural and instrumental tran-

sients. Deep networks have since been trained to detect earthquakes (Perol et al., 2018; Ross et al., 2018; Mousavi et al., 2019b) and to pick phases (Zhu and Beroza, 2019; Mousavi et al., 2020), and CRED (Mousavi et al., 2019b), which passes convolutional features to a bidirectional recurrent layer, is the closest precedent of the detector evaluated here. Standardized datasets such as STEAD (Mousavi et al., 2019a) and the models and datasets unified by SeisBench (Woollam et al., 2022) have made evaluation on curated windows predominant. Learned pickers do run on continuous data (Mousavi et al., 2020; Yeck et al., 2021), learned classifiers have been trained to separate earthquakes from impulsive noise for early warning (Meier et al., 2019), and pickers transfer well between regions at similar distances (Münchmeyer et al., 2022). A comparison of detectors at a fixed alarm budget on continuous records, with false alarms measured on catalog-quiet data, remains rare. Learned methods also address other stages of warning, from shaking prediction (Münchmeyer et al., 2021) to magnitude (Mousavi and Beroza, 2020) and detection from GNSS velocities (Başar and Çelik, 2026), which lie outside the scope of this study.

ROC-AUC is insensitive to class prevalence (Fawcett, 2006). It therefore does not reflect the number of false positives when negatives vastly outnumber positives, which precision–recall analysis (Davis and Goadrich, 2006; Saito and Rehmsmeier, 2015) and partial areas at low false positive rates (McClish, 1989) address. In seismic detection the operating region is extreme. A detector that scores a 6 s window every 0.5 s makes 172 800 decisions per station and day, and one false alarm per day then corresponds to a false positive rate below 10^{-5}. Measuring how often an alarm sequence matches a catalog by chance resembles the time-slide technique of gravitational-wave analysis, which shifts records by lags longer than any propagation time to measure accidental coincidences (Was et al., 2010). Low-cost warning networks (Wu, 2015; Kong et al., 2016; Anthony et al., 2019) and learned detectors on edge devices (Huang et al., 2020; Bassetti and Panizzi, 2021) motivate the self-contained implementation.

3. Material and methods

3.1. Data

The detector was trained on 6 s three-component windows from 181 stations of the KO (Kandilli Observatory and Earthquake Research Institute (KOERI), Boğaziçi University, 1971), 6G (Kandilli Observatory and Earthquake Research Institute (KOERI), Boğaziçi University, 2013b) and IJ (Kandilli Observatory and Earthquake Research Institute (KOERI), Boğaziçi University, 2013a) networks, retrieved from the FDSN service of the Kandilli Observatory and Earthquake Research Institute (KOERI) around cataloged events of $M \geq 2$. Earthquake windows start 2 s before the IASP91 P arrival, noise windows come from records 3 h before each event, and the two classes are balanced 1:1. The set was partitioned by station into 120, 27 and 34 stations (76 494, 18 830 and 15 812 windows), so that no station appears in two of the training, validation and test sets (Section S19 of the supplement).

The evaluation uses the continuous archive of eleven broadband stations of the Turkish National Seismic Network

(TU; Disaster and Emergency Management Authority (AFAD), 1990), operated by the Disaster and Emergency Management Authority (AFAD) in the Marmara and northern Aegean regions, 40 to 453 km apart and absent from the training set; two further stations, ARNA and CATL, are in the archive only around the Marmara replay. Events are scored only at stations where they are visible, with the signal-to-noise ratio of the 2–20 Hz vertical component around the IASP91 P arrival t_P (times in seconds),

$$\mathrm{SNR} = \mathrm{RMS}\big(x[t_P - 1,\ t_P + 12]\big)/\mathrm{RMS}\big(x[t_P - 60,\ t_P - 10]\big), \tag{1}$$

of at least 3. Of the cataloged events within 500 km, 30 438 reach this level at one or more stations.

The pipeline is evaluated on replays of the archive (Table 1). Two event windows tune the trigger settings, and two held-out sequences test them: the 2025 Sındırgı M_w 6.1 sequence and the 2026 Simav M_w 4.8 sequence, the second chosen and its protocol committed before its data were extracted. Twenty-three quiet windows with no cataloged event within 250 km of any station measure the false-alarm rate. Three public KOERI datasets, attached to the pipeline release, allow the principal results to be reproduced without the AFAD archive. The catalog is incomplete for small events during aftershock sequences, so that an alarm without a catalog counterpart is not necessarily false, and the tuning and first held-out windows belong to the same sequence.

Table 1
Replay datasets.

Dataset	Network	Start (UTC)	Duration	Stations	Catalog events	Role
Sındırgı	TU	2025-11-10 18:04:04	30 min	3	9	tuning
Marmara Sea	TU	2025-04-23 08:29:01	3 h	8	57	tuning
Sındırgı M_w 6.1	TU	2025-08-10 16:00	3 h	6	66	held out
Simav M_w 4.8	TU	2026-04-11 14:00	3 h	6	56	second held out
Quiet windows (23)	TU	2024-05-22 to 2025-03-07	102.8 h	7–8	0	false-alarm rate
Quiet window	KO	2024-01-03 11:25:00	5.8 h	6	0	false-alarm rate, public
Sındırgı M_w 4.9	KO	2025-11-10 18:04:00	31 min	6	9	public reproduction
Marmara M_w 6.2	KO	2025-04-23 09:00:00	1 h 45 min	8	37	public reproduction

Note. Catalog events are counted within 250 km of station centroid. Held-out replays were not consulted for any choice of settings. Stations of each replay are listed in Table S1; of 15 KO stations, 11 contributed training or validation windows.

3.2. Detectors

Each 6 s window (600×3 samples at 100 Hz) is detrended, tapered, band-passed at 1–45 Hz with a zero-phase filter applied to the window, whose last sample is the current time, so that no later sample is used, unlike the look-ahead mode of Section 4.3, standardized per component, and compressed with an inverse hyperbolic sine. The pipeline's detector

standardizes each window by its own statistics, $\tilde{x}_c[n] = (x_c[n] - \mu_c)/\sigma_c$, which removes gain and site amplitude, whereas the offline networks of Section 4.6 use the long-term noise statistics of the station (station scaling). Three strided convolutional layers reduce the window to 75 steps of 96 channels, and a bidirectional LSTM with 48 units per direction, a four-head self-attention block with residual connection (Vaswani et al., 2017) and a small perceptron produce a single logit (142 059 parameters). The arrangement follows the one-dimensional channel of 1D2D-EDL (Wang and Zhao, 2025). The network was trained with label smoothing (targets 0.1 and 0.9), three seeds are averaged, and the ensemble reaches a ROC-AUC of 0.9785 on the held-out curated test set. A P/S picker, run only after a trigger, provides picks for location (Section S19).

The recursive STA/LTA band-passes the trace with fourth-order Butterworth sections, forms the squared vertical component or the three-component energy, and agrees with SciPy (Virtanen et al., 2020) and ObsPy (Beyreuther et al., 2010; Krischer et al., 2015) to within rounding (Section S2). FilterPicker takes the maximum over a bank of band-pass characteristic functions, triggers at a first threshold and declares a pick when its mean over a window T_{up} reaches a second; the pipeline's port of Lomax's C code (FilterPicker5 v5.0.3) gives identical picks (Section S12). Either classical detector can replace the network, with every trigger, at most one per 15 s per station, giving P at the onset and everything downstream unchanged.

3.3. Real-time pipeline

The pipeline performs ingestion, conditioning, detection, picking, association and grid-search location (Figure 1). Each station is served by an ingestion and a processing thread that feed one network stage through an ordered message bus, and the build is one binary and a directory of weights. Records are released when the replay clock reaches the time of their last sample, which reproduces the arrival order of a live feed. Samples are held in a ring of 2^{17} samples per component, with gaps written as marker values, and because a sample's time is derived from its position, writes never drop or wait and flow control is applied at the source. A replay across a 35-minute network-wide outage reproduces exactly the alarms of the same data split around it (Section S2). A magnitude regressor is part of the pipeline but was disabled for every result here.

Windows are scored on an absolute grid every 0.5 s. Label smoothing makes the detector's outputs saturate between 0.90 and 0.907, so that a single threshold at $p \geq 0.9$, the rule first used, depends on the third decimal; a logit threshold selects the same windows, since the sigmoid is monotone. The adopted rule has a persistence branch, which fires when eight consecutive windows reach $p \geq 0.8$, 3.5 s after the first, and an instant branch, which fires when one window reaches $p \geq 0.9$. A trigger re-arms after two windows below 0.3, and no window within 15 s of the previous trigger may start a run. The P time of a learned trigger is the first sample in its window at which a recursive STA/LTA (2–20 Hz, 1 and 30 s, vertical) running alongside reaches a ratio of 3, otherwise 3.5 s into the window. This sample always lies in

the past, and the anchoring adds no latency. Two detections are associated when their P times differ by no more than the P travel time between the stations plus 3 s, an event is declared when two stations agree, and later detections at a station within 40 s are absorbed as its coda. Location, a two-stage grid search in a half-space from accepted P and S picks, serves the report after the alarm and is not evaluated (Section S16).

All layers are implemented by hand and load the PyTorch (Paszke et al., 2019) parameter names unchanged, and agreement tests compare every stage with PyTorch, SciPy, torchaudio and ObsPy on real waveforms. The worst-case error of the detector logit is 6.0×10^{-7}, and repeated and concurrent replays produce byte-identical reports. On a Ryzen 5 5600, a window of the three-model ensemble costs 5.2 ms against 0.0005 ms for the STA/LTA over the same 0.5 s, and one thread sustains 95 stations. The static aarch64 build raises the same alarms at the same times on a Raspberry Pi 4 (Cortex-A72, 1.8 GHz), where an ensemble window costs 36.3 ms and four cores sustain 52 stations (Table S4).

3.4. Evaluation protocols

For the continuous-record ranking, records were cut into disjoint windows. Windows from 10 s before to 60 s after the P arrival of any cataloged event within 500 km form its guard, and the rest form the background. The threshold for a budget of B false alarms per day is the score that B background windows a day exceed, so that detectors are compared at equal false-alarm rates, and an event counts as detected when the highest score in its guard exceeds the threshold.

On the replays, trigger settings for every detector were selected on the two tuning replays by rules stated in advance, and neither held-out sequence was used to choose any setting. The learned detector's trigger rule and the STA/LTA selection rules are recorded in the same commit as their first held-out results, and every later protocol and selection was committed before the runs it governs (Table S13). A declared event matches a catalog event if its epicenter lies within 30 km and its origin within 5 s, or if the predicted P arrival at a station that raised the alarm lies within 6 s of that station's P time. The chance level c is the detection rate with every catalog time shifted by $\pm$127, $\pm$263 and $\pm$391 s, and with observed rate o the chance-corrected rate is

$$\hat{o} = (o - c)/(1 - c), \tag{2}$$

reported as an equivalent number of events. Inside aftershock sequences the shifts inflate c for every detector, and removing shifted events near real ones moves no paired difference by more than 0.8 events (Section S17). Window probabilities are cached, which reproduces full inference exactly and makes large trigger grids feasible. Two detectors are compared on per-event outcomes by an exact McNemar test and a paired bootstrap of the difference in chance-corrected detections (10 000 resamples within each replay), accompanied by a bootstrap over 10-minute blocks of origin time. The comparisons fixed by a committed protocol are confirmatory, and the full grids, selection sensitivities

and subgroup splits are exploratory. Alarms without a catalog counterpart are called unmatched rather than false.

The false-alarm rate was measured on 23 quiet windows, with no cataloged event within 250 km of any station during the window or 30 min on either side, replayed on seven or eight stations after a 30 min lead-in (102.8 h of scored data between May 2024 and March 2025). Every detector ran with its selected settings, and both trigger grids were also replayed on these windows. Rates are given as alarms per day with exact Poisson intervals.

4. Results

4.1. Trigger rule

Relative to the single window at $p \geq 0.9$, the adopted rule detects 39 instead of 24 of the 66 held-out events, at 7 instead of 2 unmatched alarms (Section S18). The gain lies in M2–4 events, and large events, which saturate the detector and fire the instant branch, keep their alarm times (M_{w} 6.2 at +11.0 s, M_{w} 6.1 at +17.0 s and M_{w} 4.9 at +18.0 s after origin). On the held-out replay 8, 10 and 12 windows detect 39, 29 and 25 events, so that the adopted setting lies at the edge of a steep decline. Over the full grids (Figure 2a), the learned detector's envelope lies above that of STA/LTA at every count of unmatched alarms from 1 upward except 14, where both reach 38.6. A single threshold just below the plateau ($p \geq 0.895$) performs comparably but is as sensitive to the plateau's height (Section S10).

4.2. Learned detector versus STA/LTA

STA/LTA settings were tuned over 648 combinations, and two rules fixed before the held-out run selected one setting at equal unmatched alarms (2–20 Hz, 1 and 30 s, ratio 8, vertical) and one at equal detections (2–10 Hz, 0.5 and 10 s, ratio 6, vertical). At comparable unmatched alarms (7 and 9), the learned detector detects 68% more held-out events, with 36.4 against 21.7 chance-corrected detections, a difference of 14.7 (95% interval 5.9–23.7; 7.9–22.2 with 10-minute blocks; exact McNemar $p = 0.003$). The difference lies in M2–4 events (12 against 4 of 27 M2–3 events and 21 against 14 of 32 M3–4 events), whereas STA/LTA detects all six M4–5 events and the learned detector five. At comparable detections (37.9 against 36.4), STA/LTA raises about three times the unmatched alarms, 26 against 7 held out and 63 against 26 on the tuning data (Table 2, Section S18). STA/LTA alarms 1.0–1.3 s earlier on the two largest events, because it triggers on the onset sample whereas the window detector must see the P wave inside a 6 s window.

4.3. Other detectors and a second held-out sequence

EQTransformer and PhaseNet were run through SeisBench 0.12.6 in place of the detector, each 6 s window receiving the largest model output within it, with everything downstream unchanged. In the *causal* mode the model input ends at the end of the window, as in real-time operation, and in the *look-ahead* mode it is centered on the window, which gives an offline upper bound. For each model and for FilterPicker, the weights and trigger setting were selected on the tuning replays by the two rules above, and both rules chose the INSTANCE weights (Michelini et al., 2021) for

the published models. The Simav sequence was replayed once with every setting frozen, on the six archive stations within 300 km, against 56 cataloged events, 39 of them below M2 (Table 2, Figure 3).

Table 2
Every detector at settings selected on tuning data: two held-out sequences, quiet windows and public KOERI event sets.

	Sındırgı M_w 6.1 (66)			Simav M_w 4.8 (56)			Quiet	KO (46)
Detector, rule	Corr.	Unm.	Difference	Corr.	Unm.	Difference	per day	Det./Unm.
learned	36.4	7		35.1	17		3.7	36/25
STA/LTA, equal unm.	21.7	9	+14.7 (5.9, 23.7)	29.1	14	+6.0 (−2.4, 14.3)	75.4	23/5
STA/LTA, equal det.	37.9	26	−1.5 (−9.7, 6.6)	40.9	41	−5.8 (−12.5, 0.8)	127.0	39/17
FilterPicker, equal unm.	28.8	8	+7.6 (1.4, 14.1)	33.8	18	+1.3 (−7.4, 9.8)	82.4	31/12
FilterPicker, equal det.	34.9	16	+1.5 (−5.6, 8.7)	39.0	49	−3.9 (−12.7, 4.7)	207.1	36/32
PhaseNet, both	33.3	12	+3.1 (−4.0, 10.5)	36.8	28	−1.8 (−8.2, 4.1)	3.5	32/15
EQTransformer, equal unm.	11.0	13	+25.5 (15.3, 35.4)	22.7	15	+12.4 (2.9, 21.5)	1.6	24/16
EQTransformer, equal det.	34.0	50	+2.4 (−9.1, 13.9)	32.3	55	+2.8 (−5.5, 11.1)	6.1	34/37
Look-ahead, offline only								
PhaseNet, equal unm.	42.8	23	−6.4 (−15.4, 2.3)	41.2	35	−6.1 (−13.5, 1.0)	7.2	38/23
PhaseNet, equal det.	38.8	9	−2.3 (−9.2, 4.3)	41.4	27	−6.3 (−13.1, 0.3)	3.3	32/14
EQTransformer, both	32.2	11	+4.2 (−6.7, 15.3)	33.4	25	+1.7 (−6.2, 9.7)	2.6	35/15

Note. Corr.: chance-corrected detections; Unm.: unmatched alarms; difference: learned detector's chance-corrected detections minus other's, with 95% paired bootstrap interval by events. Quiet: alarms per day on 102.8 h of AFAD quiet windows. KO: detected events and unmatched alarms on demo_ko and marmara_ko together. Settings: STA/LTA as in Section 4.2; FilterPicker windows 6/10/0.4 s with thresholds 15 and 10 (equal unm.) and 3/5/0.2 s with 4 and 10 (equal det.); published models with INSTANCE weights at threshold × windows 0.7×2 (PhaseNet), 0.8×2 and 0.3×8 (EQTransformer), and with look-ahead 0.4×8, 0.6×8 (PhaseNet) and 0.5 × 8 (EQTransformer).

On Simav the direction of the STA/LTA comparison repeats, but not its size. At comparable unmatched alarms (17 and 14) the learned detector detected 35.1 chance-corrected events against 29.1, a difference of 6.0 whose interval includes zero, and the gain again lies in M2–4 events. Most of Simav's events are below M2, the lower limit of the detector's training events, where no detector separates reliably from noise. Pooled over both sequences, with the chance correction and bootstrap within each, the difference is 20.7 (8.8–32.8; exact McNemar $p = 0.001$), a summary that neither protocol specified.

The gain over STA/LTA is not specific to the pipeline's detector. Causal PhaseNet matched it on both sequences (pooled difference 1.4, −8.1 to 11.0), with 12 and 28 unmatched alarms against 7 and 17 and with 15 quiet-window alarms against 16, whereas causal EQTransformer was not competitive at equal unmatched alarms. With look-ahead, PhaseNet detected the most events of any detector, but only offline and at three times the unmatched alarms and twice the quiet-window rate of the learned detector (7.2 against 3.7 per day, exact $p = 0.04$). This gap does not come from scoring the window at the edge of the model input. Under a protocol committed before any of its scores, ending the input 1 or 2 s after the window left PhaseNet level with the learned detector and EQTransformer far below it (Section S15). FilterPicker is the strongest classical detector tested. At equal unmatched alarms it detected fewer held-out events than the learned detector (difference 7.6, 1.4 to 14.1; McNemar $p = 0.021$) and as many on Simav, and at equal detections it matched the learned detector with two to three times its unmatched alarms. On the quiet windows, however, its two settings raised 82.4 and 207.1 alarms per day, 22 and 55 times the learned detector's rate, mostly at two stations, CMH

and KIRK.

4.4. False alarms on quiet data

Over 102.8 h without a cataloged event within 250 km, the learned detector raised 16 alarms, 3.74 per day (95% interval 2.14–6.07), against 323, 75.4 per day (67.4–84.1), for STA/LTA at equal unmatched alarms (rate ratio 0.050, 0.028–0.082). That ratio of about 20 belongs to the two selected settings, because STA/LTA was matched on unmatched rather than false alarms. Over the full grids (Figure 2b), at equal tuning detections, the quietest STA/LTA setting raises 5 to 29 times the alarms of the quietest learned setting (4.9 against 0.93 per day at 30 chance-corrected detections, 14.7 against 1.63 at 44 and 60.5 against 2.1 at 48.7). The rate of STA/LTA is similar in the held-out event window (72 per day) and on quiet data, whereas the learned detector's falls from 56 to 3.7 per day, which suggests that most of its unmatched alarms in the event windows were uncataloged aftershocks. On 5.8 h of public KOERI quiet data the learned detector raised 1 alarm against 25 for STA/LTA.

Table 3
Alarms of learned detector on AFAD quiet windows (102.8 h), by criteria that need no judgment of waveforms.

Class	Alarms	Criterion
KIRK's instrumental pulse	9	KIRK declared; its detection in pulse slot
seismic, picker-confirmed	4	S−P test with PhaseNet or EQTransformer picks
undetermined	3	neither
all alarms	16	3.74 (2.14–6.07) per day
excluding picker-confirmed	12	2.80 (1.45–4.89) per day
KIRK's pulse only	9	2.10 (0.96–3.99) per day
Sensitivity: picker rule on any associated pair of stations		
pulse / seismic / undetermined	8 / 5 / 3	alarm 1 confirmed through BAND–CMH
excluding picker-confirmed	11	2.57 (1.28–4.60) per day

Note. Upper rows: pre-registered classification; last two rows: same picker rule on every pair of stations associated with an alarm, also specified before it was run. Rates: alarms per day with exact Poisson 95% intervals. Each alarm is listed in Table S11.

The 16 alarms were classified by criteria that need no judgment of the waveforms (Table 3). Nine were declared with KIRK, whose detection fell 58.6–58.8 s after a 10-minute mark, the slot of a periodic instrumental pulse. Detections associate within the P travel time between their stations plus 3 s, 14 to 67 s for the pairs with KIRK, so a faulty station raises the network's alarm rate through chance coincidences with the other stations' triggers. A rule fixed before it was run confirms a seismic source when a published picker finds P and S at two declaring stations and the two S−P distances fit the station separation and the P-time difference. It confirms four alarms of the learned detector, one an M2.4 listed by the European-Mediterranean Seismological Centre but not by AFAD, and three of the 323 STA/LTA alarms, all among the same events. Without them the rates are 2.80 and 74.7 per day, and under a classification that is the same for both detectors the contrast is unchanged. KIRK's pulse, confined to May–June 2024 and absent from the event replays, is a single 0.3 s pulse locked to UTC (Figure 4), and without KIRK the learned detector's rate would be

1.40 per day. Of the 323 STA/LTA alarms, 263 rest on signals shorter than 3 s at both stations, most of them spikes under 0.5 s, such as the isolated spikes at CMH that the learned detector ignores.

4.5. Alert latency

For the Marmara M_w 6.2, the alarm came 11.0 s after origin, and the S wave reached station ELBA (38 km) 2.1 s later. Because the alarm carries no magnitude or shaking estimate, such margins bound the lead time a warning could have had. Because stations are processed independently, a recorded replay can be re-run through the network stage for every subset of stations. Over every subset that declared the M_w 6.2, the alert exceeds the P travel time to the second-nearest station by a median of 2.05 s (interquartile range 0.6–2.8 s), and two stations alone, ARNA and ELBA, give the same alert as all eight (Sections S5–S7).

4.6. Detector ranking on a continuous record

Six detectors were scored on a continuous 728-day record at station MANT: the pipeline's learned detector and STA/LTA, and four offline detectors, a classical STA/LTA (Allen, 1978), two P-wave detectors that score 3.4 s windows and differ only in their noise windows, and a 6 s detector with station scaling (Table 4, Figure 5). The pipeline's two detectors are almost indistinguishable by event AUC (0.9910 for STA/LTA, 0.9892 for the learned detector; difference 0.0016, 0.0000–0.0035), yet at one false alarm per day the learned detector finds 34.8% of the events against 22.2%, 1.6 times as many. Among the offline detectors, recall at one false alarm per day is in the exact reverse order of event AUC. Because all six are scored on the same record, the inversion is a property of the metric, which integrates over all false positive rates whereas operation uses rates of order 10^{-4}. The guard over which an event is scored extends 60 s past P, so that every detector can also score the S wave. Scored only from windows that end at least 1 s before the predicted S, over the 12 206 events with S−P of at least 7 s, the inversion at one false alarm per day remains (33.8% against 21.4%), although STA/LTA then draws level or ahead at ten per day (Section S14).

Table 4
Recall of 13 056 events with SNR ≥ 3 over 728 days at MANT at three alarm budgets (false alarms per day).

Detector	Event AUC	100/day	10/day	1/day
Run by the pipeline				
STA/LTA (causal, 2–20 Hz)	0.9910	0.991	0.730	0.222
6 s detector (per-window)	0.9892	0.979	0.743	0.348
Offline				
STA/LTA (classical)	0.9795	0.928	0.548	0.132
P-wave, amplitude-matched	0.9622	0.903	0.627	0.183
P-wave, natural negatives	0.9516	0.861	0.573	0.219
6 s detector (station scaling)	0.9403	0.864	0.741	0.316

Note. Event AUC: maximum score within each event's guard window against maximum score over random stretches of equal length, away from cataloged events, of same record.

5. Discussion

A detector for early warning is best evaluated at the operating point at which it will be used and on the data it will see. Matched on unmatched alarms and replayed on quiet data, the learned detector and the classical triggers separate clearly, whereas event-level ROC-AUC barely separates the pipeline's two detectors and ranks four offline ones in the reverse order of their recall.

The classical detectors are neither obsolete nor superior. The learned detector gains in M2–4 events that an amplitude criterion cannot separate from noise, and it keeps that sensitivity at a small fraction of the classical detectors' alarm rate on quiet data. The classical detectors, however, alarm 1.0–1.3 s earlier on the largest events, cost orders of magnitude less, as measured for STA/LTA, and time the onset to a sample, which is why the pipeline uses STA/LTA for timing rather than for decisions. Causal PhaseNet, trained on another country's data, matched the learned detector, and no published model that can run in real time was shown to exceed it at an alarm rate matched on the tuning data. The gain thus comes from learned detection rather than from this particular detector, and what distinguishes the pipeline's detector is that it reaches this level in a self-contained program with 142 059 parameters per network.

The computational design carries much of the protocol's weight. Cached window probabilities and deterministic replays made trigger grids of hundreds of settings, seven scoring passes per setting and pre-registered evaluations practical, and replaying every station subset separated the geometric term of the alert time from the terms the detector controls. The P travel time to the second station dominates the alert, and the detector contributes about 2 s after it, so that stations near source regions shorten the alert more than a faster detector would. Two failures of the data path were found only by complete replays (Section S2). The dataloggers nearest the M_{w} 6.2 switched to uncompressed integers, which a decoder that accepted only Steim-2 silently rejected, and an earlier version that dropped samples under load shifted later samples in time and lost 15 of 44 detections.

5.1. Limitations

The study rests on one regional network and two held-out sequences, the second with an advantage over STA/LTA that was not significant, and the first scored repeatedly for post hoc analyses. The ranking on a continuous record rests on one station. The catalog is incomplete for small events, so that unmatched alarms only bound false alarms, and absorbing detections within 40 s as coda trades the resolution of closely spaced events, a doublet among them, for fewer false alarms in dense sequences, and the quiet-window classes use no analyst picks. The published models ran with pre-trained weights and the pipeline's trigger rule, and fine-tuning them on the detector's training split might improve them. The input is a replay of archived files, and a live SeedLink client, with its latency, packet loss and clock problems, is not implemented. Further held-out sequences committed before extraction, more stations and a live shadow run beside AFAD's alerts would address these limits.

6. Conclusions

A self-contained, verified real-time pipeline and a replay protocol with chance correction and pre-registered held-out sequences measure seismic detectors at their operating point. Under this protocol, a learned detector found 36.4 chance-corrected events of 66 against 21.7 for STA/LTA at comparable unmatched alarms on one held-out sequence and 35.1 of 56 against 29.1, within the uncertainty, on a second. On quiet data it raised a small fraction of the classical detectors' alarms, and causal PhaseNet matched it. Two detectors with nearly equal event AUC differed by a factor of 1.6 in recall at one false alarm per day, and detectors for early warning are therefore best compared on continuous records at fixed alarm budgets.

Code availability

Name of the software: ayzek. Developer: H. Oğuz Bolat (oguz.bolat@std.yildiz.edu.tr), first available in 2026. Program language: C++23, with Python tools for model export and evaluation. Hardware requirements: an x86-64 or aarch64 processor; tested on a desktop computer and on a Raspberry Pi 4. Software required: Meson and Ninja and a C++23 compiler (GCC 16 or Clang); no machine-learning runtime. Program size: about 9 400 lines of C++ and 6.5 MB of model weights. License: GNU General Public License, version 3 or later. The source code, model weights, documentation and the scripts that reproduce every result (`tools/reproduce_all.sh`, with expected values in `bench/claims.json`) are available at `https://github.com/hogib/ayzek` (release v0.1.0-beta.6), archived at `https://doi.org/10.5281/zenodo.23672495`. The FilterPicker port is derived from the C code of Lomax et al. (2012) (GNU GPL version 2 or later). Software versions are listed in Table S9.

Data availability

Waveforms are from the Turkish National Seismic Network (Disaster and Emergency Management Authority (AFAD), 1990), obtained through the AFAD archive service, and from the KO network of KOERI (Kandilli Observatory and Earthquake Research Institute (KOERI), Boğaziçi University, 1971) through its FDSN service; the AFAD and KOERI catalogs are public. The AFAD waveforms cannot be redistributed. The three KOERI datasets of Table 1, which reproduce the principal results, are attached to the pipeline's release. The offline evaluation code is available at `https://github.com/SismoKaos-Research/cnn_earthquake`, the training code of the detector at `https://github.com/hogib/cascade` and that of the picker at `https://github.com/hogib/sphase`.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude Opus 5 and Claude Opus 5.5 (Anthropic) to develop and refactor parts of the pipeline code, extend the analysis and edit the manuscript text. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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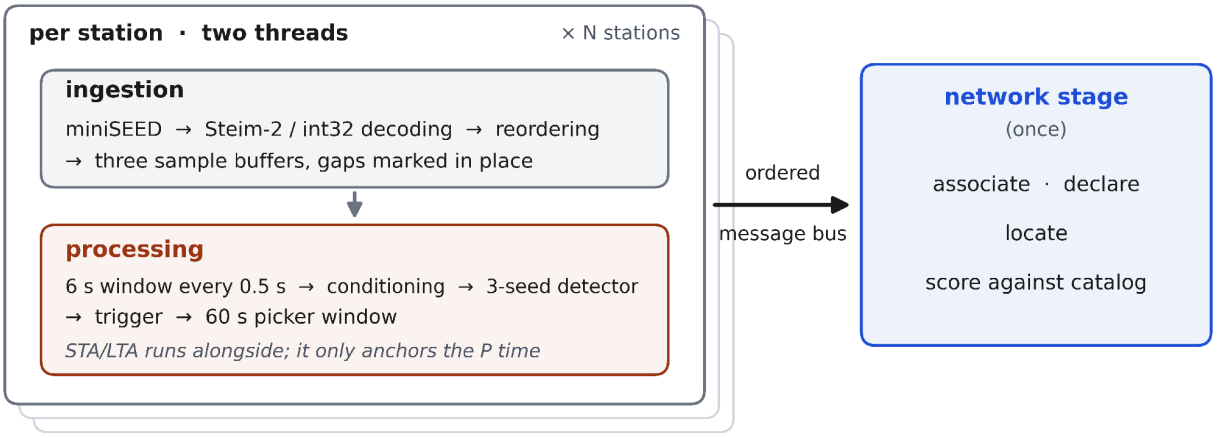


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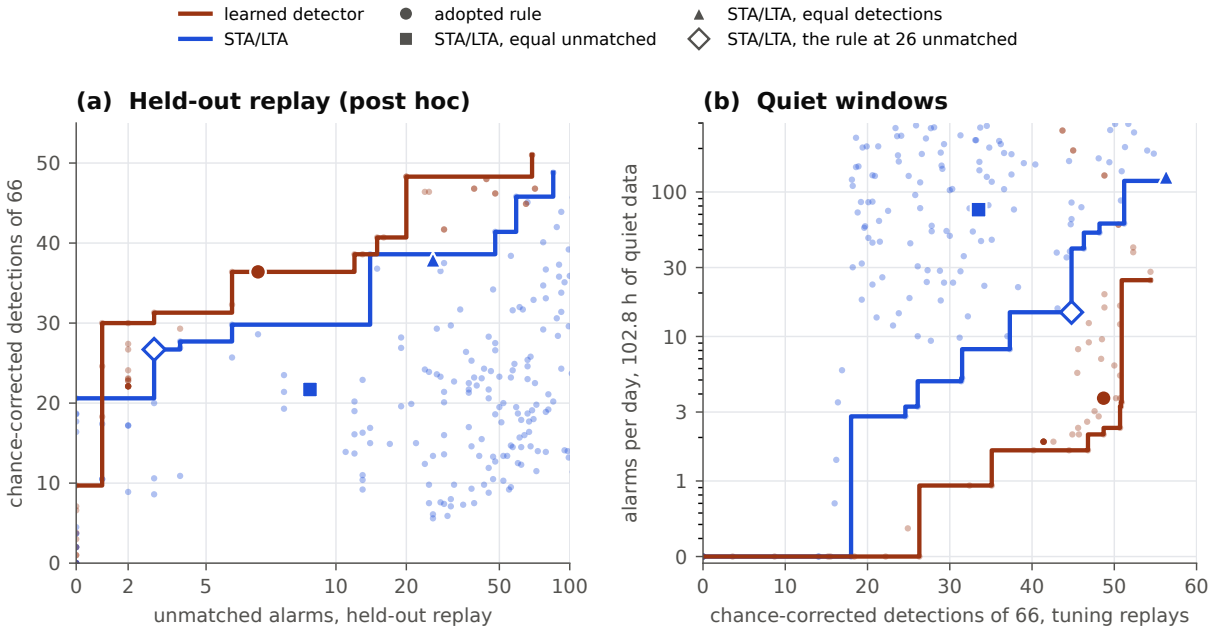


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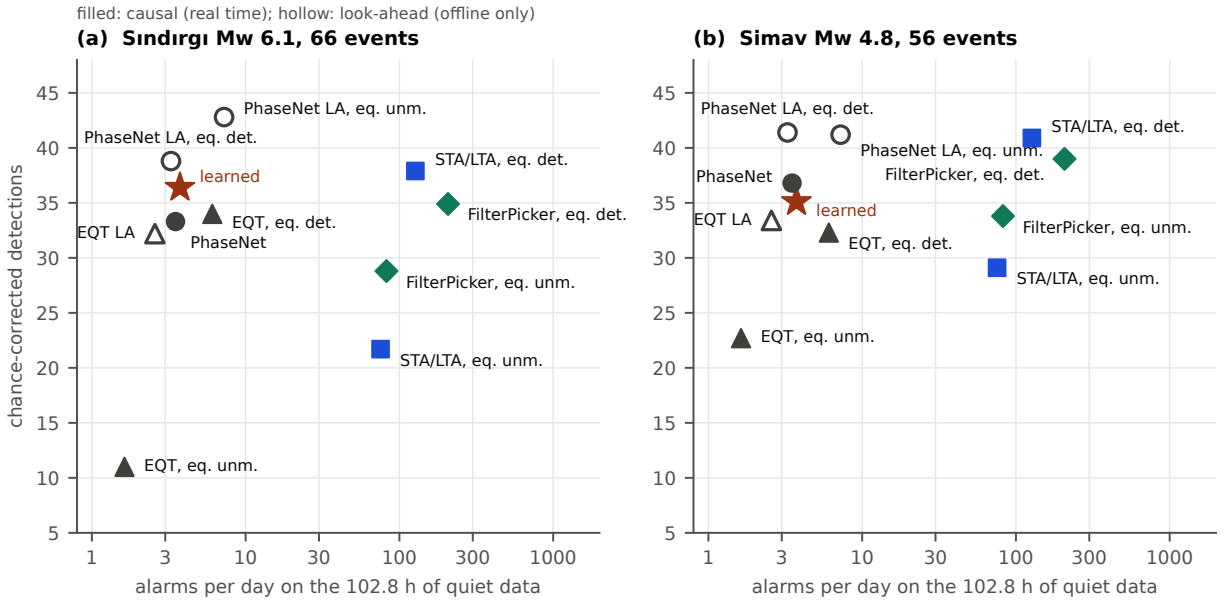


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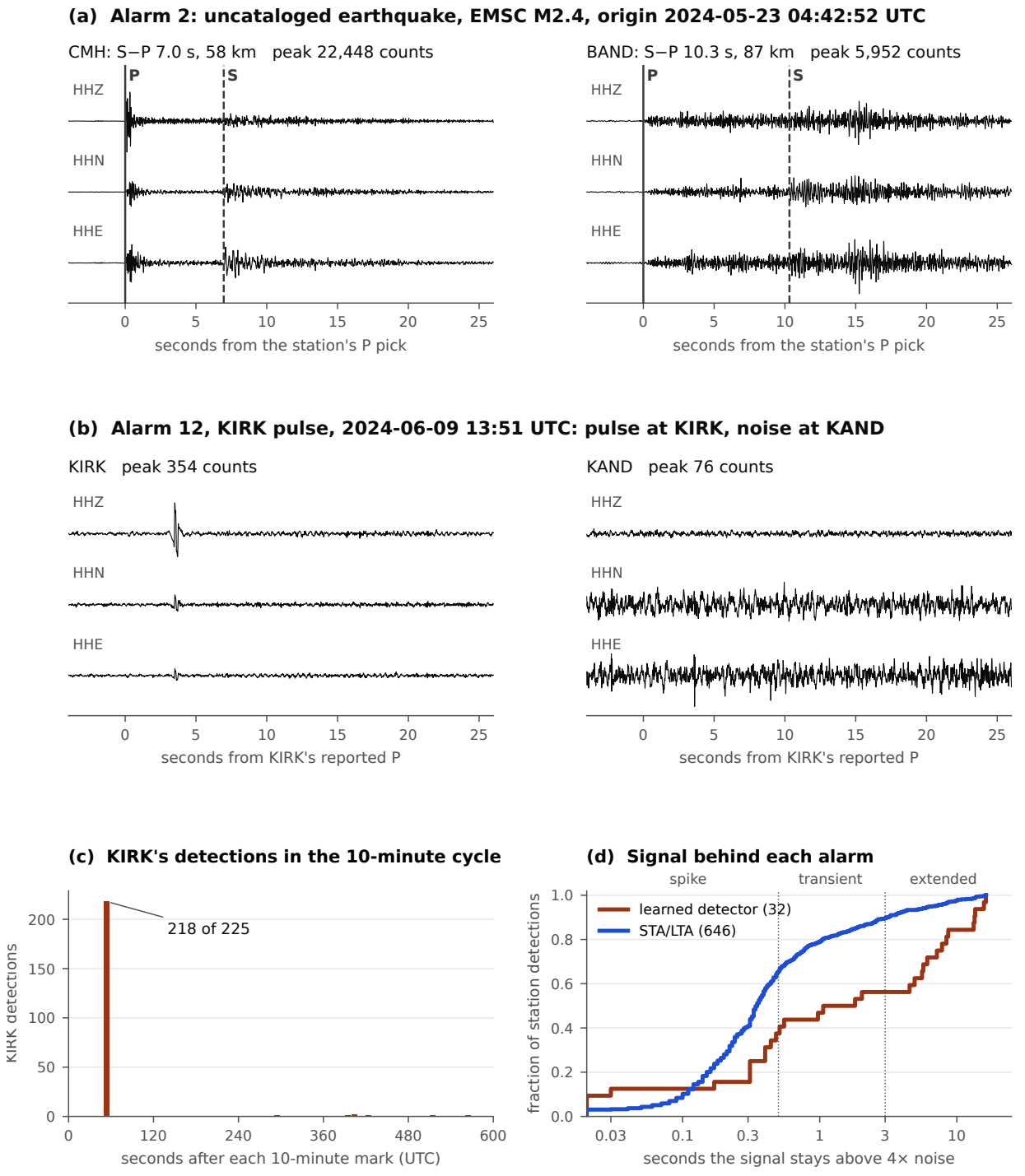


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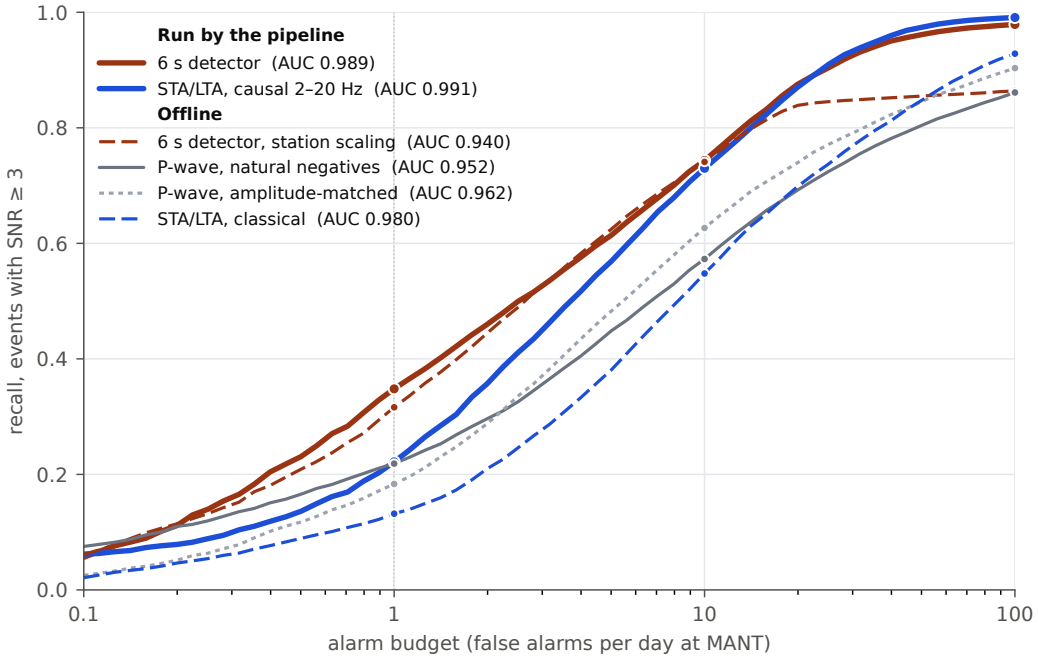


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